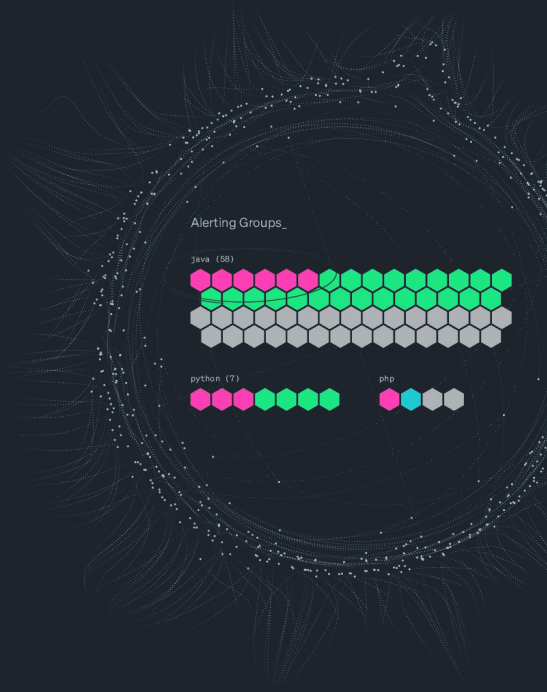


New Relic Alerts Fundamentals

Phil Weber
Technical Instructor

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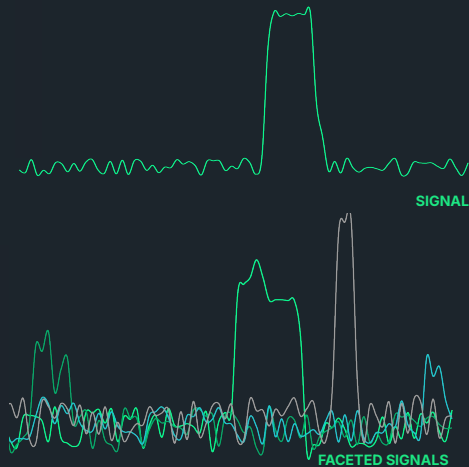


Agenda

- 01 Alerts terminology and concepts
- 02 Lab: Creating alert conditions and policies
- 03 Lab: Configuring workflows and destinations
- 04 Muting rules
- 05 Managing issues
- 06 Additional resources

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What is a "Signal"?



Acquisition Detection Noise reduction Routing Notify

Signal

- **Any data** being sent to New Relic
- Result of an **NRQL query** against incoming data
- If you can **query it you can alert** on it!

Faceted signal

- Faceting the query results in **multiple signals**
- Each signal can trigger separate alerts

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Signals to alert on

Some ideas:

- Host metrics: CPU, memory, disk space, restarts
- APM golden metrics: Latency, throughput, errors
- Key transactions
- Synthetic journeys: Success rate, duration
- Log data: Error messages, business data
- Business metrics: Customers, sales

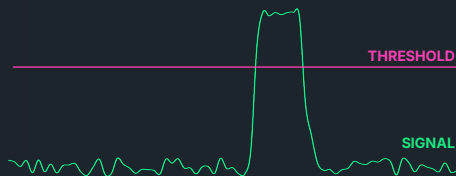


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Conditions

Signal acquisition and evaluation



Some examples

Signal	Threshold
% Error rate	% error > 5%
Average basket value	median(value) < \$20
Login attempts per user	uniqueCount(user) > 10

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[Documentation on alert conditions](#)

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Acquisition

Detection

Noise reduction

Routing

Notify

Signal query

Defined by NRQL query including optional WHERE and FACET clauses.

Guided wizard available to help build.

**Additional non-NRQL conditions soon to be deprecated*

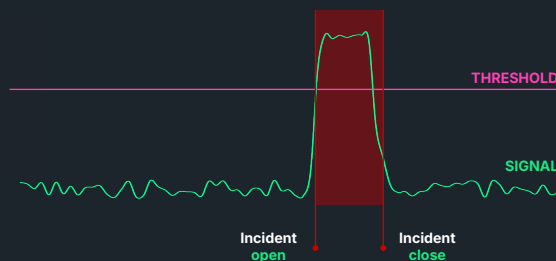
Threshold

Value, which if breached by signal for a defined time period, will trigger an incident to be opened.

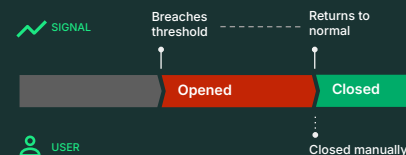
A threshold can be set for both **warning** and **critical**

Incidents

Signal breach detection



Incident lifecycle



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Acquisition

Detection

Noise reduction

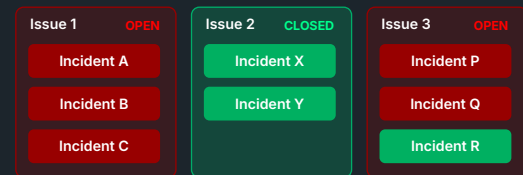
Routing

Notify

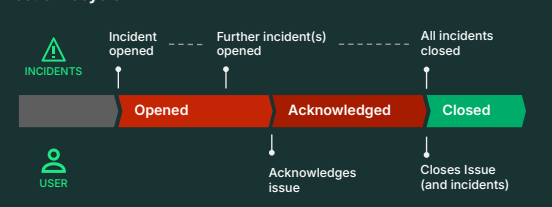
- Incidents **open** when a **signal breaches the threshold** defined in a condition.
- Incidents **close automatically** when the signal is no longer in breach.
- There will be some **latency** between signal breaching and incident opening based on condition settings.
- Incident will open for **each signal facet**
- Incidents can be **manually closed** by user

Issues

Incident management and noise reduction



Issue lifecycle



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[Documentation about Issues](#)

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Acquisition Detection Noise reduction Routing Notify

Incidents are the symptoms of a larger problem (the issue).

Issues **group incidents together**, reducing noise and driving notification workflows.

- Issues are **opened when incidents open**
- Issues can **contain multiple incidents**
based on policy preference or correlation decisions
- Issues **close automatically** when all contained incidents have closed or if inactive for defined period
- Issues can be **manually closed**, which close all contained incidents.

Policies

Predetermined incident grouping

FEWER
ISSUES

1

Per
Policy



Only one issue will be open at a time for the **entire policy**.

- Requires immediate action and closing the issues to be effective

2

Per
Condition



One issue will be open at a time for **each condition** in your policy.

- Useful for policies containing conditions that focus on entities that perform the same job

3

Per
Incident



An issue will be created for **every incident** of **each condition** in your policy.

- Useful if you need to be notified of every violation or if you have an external system where you want to send alert notifications

MORE
ISSUES

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[Documentation on alert policies / issue preference](#)

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Acquisition Detection Noise reduction Routing Notify

Lab: Alert conditions and policy

Perform the following tasks in NRU Training account:

1. Go to *Alerts > Alert Conditions* and create two alert conditions:
 - Create a critical incident if any APM service in the *Production* environment has a response time greater than 500 ms for at least 5 minutes
 - Create a critical incident if the page load time for the New Relic Pet Clinic Browser application is greater than 2 seconds for at least 5 minutes
2. Add the two conditions to a new alert policy. Give it a name you will recognize, such as *[Your Name] Policy*

Lab: NRQL alert conditions

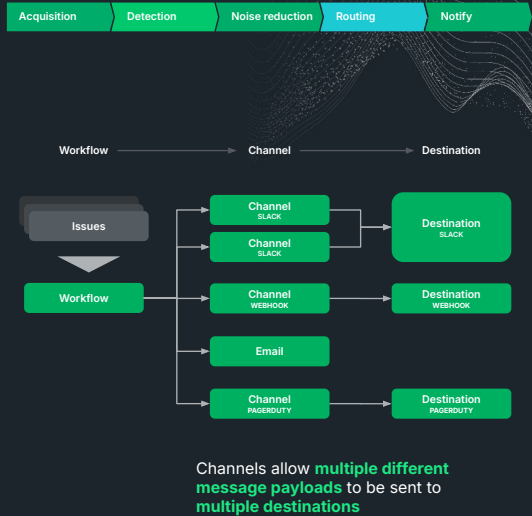
Perform the following tasks in NRU Training account:

1. Build a query that returns the average CPU utilization for any host running the *New Relic Pet Clinic* APM application
2. Create a *critical* incident if the query returns a value greater than 50% for at least 2 minutes
3. *Optionally*, add a threshold that creates a *warning* incident if CPU utilization is greater than 20% for at least 2 minutes
4. Add the new condition to the alert policy you created earlier

Workflows

Notification triage and routing

- ❑ Control **when** you want to receive notifications about issues
- ❑ **Notify correct teams** based on issue context
- ❑ Channels offer comprehensive **payload templating** options
- ❑ **Enrich notifications** with additional New Relic data



Destinations

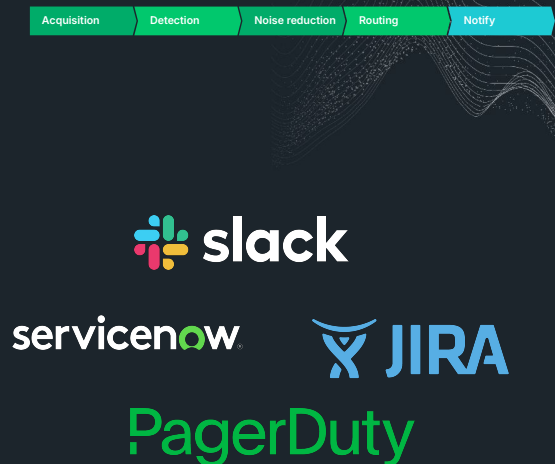
Notification dispatch

Destination inform **people** or **third party services** about issues state change.

Supported destinations:

- ❑ Email & mobile push
- ❑ Native: Jira*, ServiceNow*, PagerDuty*, Slack
- ❑ Webhook: e.g. OpsGenie, MS Teams, etc..
- ❑ AWS EventBridge

Destinations **setup once per account** and can be used by multiple workflows.



*two way integrations

Lab: Workflows and destinations

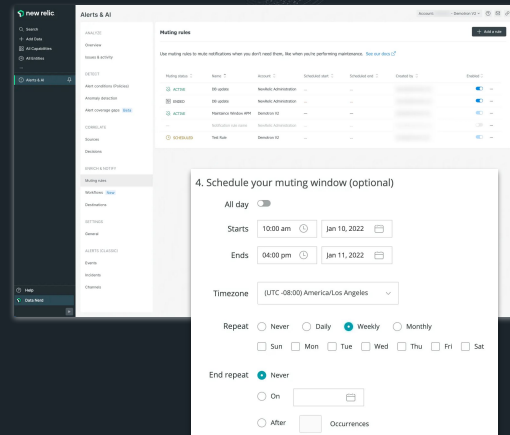
Perform the following tasks in **NRU Training** account:

1. Select **Alerts > Workflows** and add a workflow
2. Give it a name you will recognize, such as *[Your name] workflow*
3. Under **Filter data**, create a Basic or Advanced filter that matches your high CPU condition
4. Under **Notify**, select *Email* and enter your email address
5. *Optional*: Click *Test workflow*. Did you receive an email?
6. Click the *Activate workflow* button in the lower-right corner

Muting rules

Temporarily mute notifications

- ❑ Mute **issue notifications**, for instance when **performing maintenance** or known **down periods**
- ❑ Issues are **still generated and queryable**, they just don't send notifications.
- ❑ **Filter incidents** by attribute, can target multiple workflows and destinations.
- ❑ Muting rules can be **scheduled and repeatable**



Resources

- [Alerts documentation](#)

Blog posts:

- [Best practices for fixing your alerts](#)
- [Alert! Best practices for alert quality](#)

Documentation links:

- [Alert conditions](#)
- [Issues and incident management](#)
- [Alert policies](#)
- [Issue incident grouping preference](#)
- [Workflows in New Relic](#)
- [Notification destination integrations](#)



Data Bytes

Short how-to videos covering a variety of topics

- [The new alert condition creation form](#)
- [Never miss an alert with New Relic's mobile app](#)
- [Setting muting rules for workflows](#)
- [How to set up webhook destinations for workflows](#)
- [Set up sliding window aggregation in your alert conditions](#)
- [Alert aggregation methods explained](#)

Destination setup guides

- [How to configure workflow email destinations](#)
- [Set up a JIRA workflow destination](#)
- [Send notifications to Slack using workflow destinations](#)
- [How to configure workflow Pagerduty destinations](#)
- [How to configure ServiceNow workflow destinations](#)